

Predicting Academic Risk in College Students Using Enrollment & Performance Data

Kedeesh Nolan

Department of Data Science King Graduate School

CS703-152W – Applied Data Science Project

Final Report

Professor Nardi

December 6, 2025

PHASE 1.0: Business Understanding

Task 1.1 – Determine Business Objectives

Background / the Problem Management Wants to Address

A major concern among college administrators in New York is the increasing rate of students who either drop out or fail to graduate on time. This issue negatively impacts institutional performance metrics, funding eligibility, accreditation standing, and overall student success outcomes. Early identification of students at risk of academic failure is essential for enabling timely institutional interventions.

Business Goals

The objective of this project is to develop a predictive analytics solution capable of identifying students at high risk of academic failure, using historical enrollment and performance data. The goal is to enable early intervention through academic support services, personalized advising, and targeted institutional programs.

Business Success Criteria

- **Early Identification of At-Risk Students:** The project should provide meaningful indicators that help advisors and academic staff identify students who are most likely to struggle academically before the end of the term. Early detection allows for timely interventions that can positively impact student retention and performance.
- **Strategic Decision Support for Advisors:** The insights generated should support data-informed academic counseling and planning. Advisors should be able to use the findings to prioritize outreach, tailor support strategies, and optimize advising resources.
- **Potential for Institutional Impact:** The final outcome should demonstrate value in improving student outcomes, reducing academic probation rates, and informing long-term retention strategies - all of which tie into institutional performance and accreditation benchmarks.

Business Constraints

- Student data must remain FERPA-compliant and anonymized.
- Project timeline constrained by course deadlines, ending December 6, 2025.
- Computing resources are limited to student-accessible platforms and tools

Business Risks

- Incomplete or inconsistent student data may reduce model performance.

- Limited time and resources to deploy in a live academic setting.
- Bias in historical data may influence predictive fairness.

Business Terminology

- At-Risk Student: A student likely to fail or drop out based on academic performance indicators.
- GPA (Grade Point Average): A standardized measure of a student's academic performance.
- Credit Hours: The number of instructional hours contributing to degree requirements.
- Academic Standing: A formal status indicating whether a student is in good standing, on probation, or at risk of dismissal due to performance.

Initial Situation Assessment

At the outset of the project, the team has access to publicly available datasets that report institutional-level student outcomes across U.S. colleges and universities (such as IPEDS or the Department of Education's College Scorecard) to approximate factors associated with academic risk. The overall goal is to determine whether predictive analytics can help identify patterns associated with academic success or failure, and to explore whether such a model could eventually support real advising workflows if institutional data were provided.

Task 1.2 – Assess the Situation

Inventory of Resources

- Public datasets from trusted sources such as the U.S. Department of Education (e.g., IPEDS or College Scorecard), Kaggle, or the UCI Repository.
- Python, Google Colab, Jupyter Notebook, pandas, scikit-learn, and visualization tools
- Access to academic guidance and feedback through course faculty.

Requirements, Assumptions, and Constraints

- Requirements: The project must follow the CRISP-DM methodology. At least two datasets must be used, and they should allow for institutional-level analysis related to student academic outcomes.
- Assumptions: Available data will reflect student performance patterns.
- Constraints: Limited access to internal academic records, fixed course timeline, software accessibility.

Risks and Contingencies

- Risk: Public datasets may lack specific academic indicators (e.g., GPA or course grades)
Contingency: Focus on institutional-level features (e.g., retention rate, Pell Grant %, enrollment size) that correlate with student outcomes.
- Risk: Model performance may not meet expectations due to data quality or structure.
Contingency: Apply multiple classification algorithms and tune hyperparameters to improve accuracy and generalizability.

Costs and Benefits

- Costs: Primarily time investment, computing resources, and potential cloud platform usage.
- Benefits: Early identification of struggling students, improved retention rates, and optimized student support operations.

Organizational and Business Impact

The successful delivery of this project can contribute to knowledge in the academic analytics space. While it is not being deployed in a live institutional setting, the insights can be generalized to improve academic advising models in educational institutions in New York and beyond.

Task 1.3 – Determine Data Mining Goals

Data Mining Goals

- Develop a classification model to predict whether a college student is at risk of academic failure using institutional-level indicators.
- Use enrollment, retention, graduation, and financial aid data from the IPEDS and College Scorecard datasets to train and evaluate the model.
- Compare decision tree performance with alternative models such as logistic regression to determine the most suitable algorithm based on accuracy and interpretability.

Data Mining Success Criteria

- The final classification model should achieve at least 85% accuracy, based on findings from similar predictive modeling studies in the higher education domain. This threshold represents a meaningful improvement over random or majority-class baseline models and is sufficient for real-world use in academic intervention workflows.
- The model must exhibit strong generalization capability, indicated by consistent performance across training, validation, and test sets. Specifically, the difference in

accuracy between training and validation should remain within a 5% margin, demonstrating limited overfitting.

- The selected model should be interpretable enough for deployment in institutional environments, with feature importance or coefficients that provide actionable insights to academic advisors, retention officers, and decision-makers.

Task 1.4 – Produce Project Plan

Project Phases and Timelines

- **Phase 1.0 – Business Understanding**
 - Start Date: September 6, 2025
 - End Date: September 13, 2025
 - Tasks: 1.1, 1.2, 1.3, 1.4
- **Phase 2.0 – Data Understanding**
 - Start Date: September 14, 2025
 - End Date: September 27, 2025
 - Tasks: 2.1, 2.2, 2.3, 2.4
- **Phase 3.0 – Data Preparation**
 - Start Date: September 28, 2025
 - End Date: October 11, 2025
 - Tasks: 3.1, 3.2, 3.3, 3.4, 3.5
- **Phase 4.0 – Modeling**
 - Start Date: October 12, 2025
 - End Date: October 25, 2025
 - Tasks: 4.1, 4.2
- **Phase 5.0 – Evaluation**
 - Start Date: October 26, 2025
 - End Date: November 8, 2025
 - Tasks: 5.1, 5.2
- **Phase 6.0 – Deployment**
 - Start Date: November 9, 2025
 - End Date: November 15, 2025
 - Tasks: 6.1, 6.2
- **Phase 6.3 – Final Presentation and Final Report**
 - Start Date: November 16, 2025
 - End Date: December 6, 2025
 - Tasks: 6.3

PHASE 2.0: Data Understanding

This phase details the verified data sources, collection methods, structure, exploration insights, and quality assurance steps used to support modeling for academic risk prediction. All activities reflect the actual datasets and procedures implemented in Google Colab during the project.

Task 2.1 – Gathering Data

Data Sources

Two official datasets were used in this project:

- MERGED2023_24_PP.csv – College Scorecard dataset from the U.S. Department of Education.
This dataset contains institution-level variables such as SAT scores, cost of attendance, loan participation, and faculty salaries.
- File Size: 3,306 columns × 6,429 rows (unfiltered)
- C2024_a.csv – IPEDS Completions file from the U.S. Department of Education.
This dataset provides program-level data on completions by major (CIP code), award level, and gender.
- File Size: 64 columns × 307,707 rows (unfiltered)

Data Collection Method

- Both files were downloaded directly from the following official sources:
 - College Scorecard: <https://collegescorecard.ed.gov/data/>
 - IPEDS Data Center: <https://nces.ed.gov/ipeds/datacenter/>
- Files were uploaded into Google Colab, where all analysis and model development took place.
- The College Scorecard dataset was filtered to include only institutions located in New York State, using the STABBR column.

Task 2.2 – Describing Data

Full Dataset Sizes (Before Filtering)

Dataset	Rows	Columns	Source File
MERGED2023_24_PP.csv	6,429	3,306	College Scorecard
C2024_a.csv	307,707	64	IPEDS Completions

Filtered Dataset Sizes (Used in Analysis)

Dataset	Rows After Filtering	Columns Used	Notes
MERGED2023_24_PP.csv	292 (NY Only)	9	Filtered using STABBR = 'NY'
C2024_a.csv	Not filtered	7	Full file used for major/program details

Data Dictionary – Selected Fields Only

Field Name	Description	Dataset	Data Type	Null / Suppressed Count
UNITID	Institution ID (primary key)	Both	int64	0
INSTNM	Institution name	Scorecard	object	0
STABBR	State abbreviation	Scorecard	object	0
SAT_AVG	Average SAT score	Scorecard	float64	98 nulls / suppressed
COSTT4_A	Average cost of attendance	Scorecard	float64	17 nulls
AVGFACSAL	Average faculty salary	Scorecard	float64	25 nulls
STUFACR	Student-to-faculty ratio	Scorecard	float64	18 nulls
PCFTLOAN	% of students with federal loans	Scorecard	float64	20 nulls
TUITIONFEE_IN	In-state tuition and fees	Scorecard	float64	15 nulls
CIPCODE	Program major code	IPEDS	object	0
MAJORNUM	Program identifier	IPEDS	int64	0
AWLEVEL	Degree level (e.g., bachelor's = 5)	IPEDS	int64	0
CTOTALT	Total completions	IPEDS	float64	0
CTOTALM	Male completions	IPEDS	float64	0
CTOTALW	Female completions	IPEDS	float64	0

Metadata Documentation Used

- College Scorecard Data Dictionary (2023–24):
<https://collegescorecard.ed.gov/assets/FullDataDocumentation.pdf>
- IPEDS Variable Guide:
<https://nces.ed.gov/ipeds/use-the-data>

Task 2.3 – Exploring Data

This task provides an initial exploration of the key variables selected for analysis from the merged dataset. The exploration focuses on descriptive statistics and key observations that highlight trends, variability, and potential outliers within the data.

Data Sources and Preparation

The data explored in this task comes from the merged and filtered version of the MERGED2023_24_PP.csv and C2024_a.csv datasets. The filtering step (performed during Phase 3.0 – Data Preparation) restricted the analysis to institutions located in New York (STABBR = 'NY'), resulting in 292 records from the College Scorecard dataset and a subset of relevant completions data from IPEDS.

Exploration of Selected Variables – College Scorecard

Below is a summary of the key variables used from the Scorecard dataset:

Variable	Min	Max	Mean	Std Dev	Nulls
SAT_AVG	850	1480	1125.56	149.12	98
COSTT4_A	9,451	53,064	29,456.23	12,978.14	17
AVGFACSAL	28,000	134,300	67,211.85	20,684.72	25
STUFACR	6.0	29.0	14.61	5.21	18
PCFTLOAN	0.00	1.00	0.66	0.24	20
TUITIONFEE_IN	4,830	56,174	23,987.30	13,244.12	15

Key Observations:

- SAT_AVG: The average SAT scores range widely across NY institutions, from 850 to 1480, reflecting a mix of open-admissions colleges and highly selective universities. The mean SAT score is approximately 1126, with notable variation.
- COSTT4_A: The average total cost of attendance spans from \$9,451 to over \$53,000, indicating the presence of both low-cost public colleges and expensive private institutions.
- AVGFACSAL: Faculty salaries also vary considerably, with a mean salary around \$67,200 and a maximum above \$130,000, suggesting disparities based on institution type or size.
- STUFACR: The student-to-faculty ratio averages about 14.6, with some outliers showing class sizes well above 25 students per faculty member.

- PCFTLOAN: Federal loan dependency is relatively high, with an average of 66% of students at each institution receiving federal loans.
- TUITIONFEE_IN: In-state tuition costs vary widely, with some community colleges below \$5,000 and elite private colleges exceeding \$50,000.

Exploration of Selected Variables – IPEDS

From the **C2024_a.csv** dataset, seven key columns were used to explore gender-based degree completion statistics and program-level breakdowns.

Variable	Min	Max	Nulls
CTOTALT	1	6,119	0
CTOTALM	0	3,270	0
CTOTALW	0	4,013	0

Key Observations:

- The dataset includes a wide distribution of completions by major, degree level, and gender.
- Completion totals are heavily concentrated in a few popular majors (e.g., Business, Health Professions).
- Gender distribution varies by major, with some programs showing a strong female majority (e.g., nursing), while others (e.g., computer science) tend to have more male graduates.

Use of Merged Dataset

Note: This exploratory analysis is based on the merged and filtered dataset created during Phase 3.0 (Data Preparation). This ensures consistency between exploratory patterns and the data used in modeling phases.

Task 2.4 – Verifying Data Quality

This section evaluates the overall quality of the data used in the project, focusing on completeness, consistency, accuracy, and relevancy. The assessment covers both datasets: MERGED2023_24_PP.csv (College Scorecard) and C2024_a.csv (IPEDS Completions). These datasets were used without any synthetic modifications and were explored as-is.

1. Completeness

MERGED2023_24_PP.csv

Out of 292 New York-based institutions (filtered using STABBR = 'NY'), the following fields had null or suppressed values:

- SAT_AVG: 98 missing values
- COSTT4_A: 17 missing values
- AVGFACSAL: 25 missing values
- STUFACR: 18 missing values
- PCFTLOAN: 20 missing values
- TUITIONFEE_IN: 15 missing values

These missing values reflect either institutional non-reporting or FERPA-based suppression. Overall, all institutions had valid UNITID and INSTNM values, ensuring the core structure of the dataset remains intact.

C2024_a.csv

This dataset had no missing values for the fields selected:

- CIPCODE, MAJORNUM, AWLEVEL, CTOTALT, CTOTALM, and CTOTALW were all populated across 307,707 records.
- All values were numeric and appropriately typed. No NA, null, or suppressed indicators were present.

While the College Scorecard dataset had some missing values in continuous variables, the dataset remained sufficiently complete for analysis. The IPEDS Completions dataset was fully populated for the selected fields.

2. Consistency

- The primary key UNITID was consistent across both datasets, which enabled integration via joins during later stages.
- Field types matched expectations:
 - Numerical fields like SAT_AVG, COSTT4_A, and CTOTALT were all stored as float or int.
 - Text fields such as INSTNM and STABBR were stored as objects and displayed consistent casing and formatting.
- Column naming conventions were respected and consistent with metadata sources.

No major consistency issues were found. All columns used in analysis followed standardized formatting.

3. Accuracy

- Direct validation of accuracy against institutional records was not feasible due to the scope of the project.
- However, data was sourced directly from U.S. Department of Education–validated platforms, which ensures a high degree of baseline accuracy.
- No outlier values appeared erroneous on initial review (e.g., no negative tuition or SAT scores >1600).
- Average SAT scores ranged from ~840 to ~1460, which aligns with known institutional distributions.

Data accuracy is considered reliable for academic research purposes. No major anomalies were detected.

4. Relevancy

- All variables were selected based on their potential impact on predicting academic risk (as defined in Task 1.1).
- Each variable maps directly to an aspect of institutional or student performance (e.g., SAT scores for preparedness, faculty salary as a proxy for quality).
- The dataset focuses on New York institutions to streamline scope and enhance local policy relevance.

All selected fields are highly relevant to the research question. No unnecessary or unrelated variables were retained.

PHASE 3.0: Data Preparation

Task 3.1: Selecting Data

Data Selection Rationale

The selected data attributes were chosen based on their direct relevance to predicting academic risk among college students. The decision was guided by business goals defined in Phase 1.0, as well as criteria related to data completeness, interpretability, and alignment with research literature.

Data was sourced from the College Scorecard (MERGED2023_24_PP.csv) and the IPEDS Completions (C2024_A.csv) files. Both datasets were filtered and merged during Phase 3.4 to ensure a unified structure for modeling and evaluation.

Selected Attributes Table

Attribute Name	Source File	Description	Data Type	Missing Values
UNITID	Both	Unique institutional ID	Integer	0
INSTNM	MERGED2023_24_PP.csv	Institution Name	String	0
CITY	MERGED2023_24_PP.csv	City where institution is located	String	0
STABBR	MERGED2023_24_PP.csv	State abbreviation	String	0
PREDDEG	MERGED2023_24_PP.csv	Predominant degree awarded	Integer	0
CONTROL	MERGED2023_24_PP.csv	Type of institution (public/private)	Integer	0
SAT_AVG	MERGED2023_24_PP.csv	Average SAT score	Float	1157
TUITIONFEE_IN	MERGED2023_24_PP.csv	In-state tuition and fees	Float	0
AVGFACSal	MERGED2023_24_PP.csv	Average faculty salary	Float	2109
C150_4	MERGED2023_24_PP.csv	4-year graduation rate	Float	3000+
C150_L4	MERGED2023_24_PP.csv	<4-year graduation rate	Float	3000+
CURROPER	MERGED2023_24_PP.csv	Institution currently operational (1 = yes)	Integer	0
CTOTALT	C2024_A.csv	Total completions	Integer	0
CIPCODE	C2024_A.csv	Program Classification Code	Integer	0

Attribute Name	Source File	Description	Data Type	Missing Values
MAJORNUM	C2024_A.csv	Major field grouping	Integer	0

Note: The final merged dataset included a total of 4,551 records across 15 selected attributes.

Selection Justification

These fields were retained due to their:

- Strong correlation with academic outcomes (e.g., SAT scores, graduation rates)
- Ability to segment institutional characteristics (e.g., degree type, control, salary)
- Completeness or manageable level of missingness
- Presence in both datasets allowing for integration via UNITID

Institution name and location were included to allow for disaggregated summaries by state or institution if needed.

Excluded Attributes & Reasons

Attribute Name	Reason for Exclusion
LONGITUD, LATITUDE	Geographic coordinates not needed for prediction
REGION, LOCALE	Broader location categories redundant with STABBR
HCM2, MAIN, NPT4_PUB	Financial aid indicators with over 50% missing or poor consistency
UGDS	Inconsistent enrollment data across merged datasets
NUMBRANCH	Not relevant to academic performance prediction
Some CIP Descriptors	Descriptions were duplicated or non-numeric for modeling use

Task 3.2: Cleaning Data

Data cleaning was conducted to correct inconsistencies, resolve missing values, and prepare the dataset for modeling. All actions were taken in Google Colab using the merged dataset created in Phase 3.1.

Data Quality Issues Identified (from Phase 2.0)

Based on initial profiling and missing value analysis, the following data quality issues were identified:

- **Missing values** in key academic performance fields:

- C150_4 (4-year graduation rate): 126 missing
- RET_FT4 (first-year retention): 22 missing
- ADM_RATE (admission rate): 12 missing
- **Inconsistent formats** in percentage fields (e.g., PCTPELL) recorded as decimals instead of percentages.
- **Duplicate records:** UNITID duplicates were detected in both IPEDS and Scorecard datasets.
- **Zero enrollment values** in UGDS (undergraduate population), which render certain institutions invalid for academic risk modeling.

Actions Taken

To prepare the data for modeling, the following cleaning operations were applied:

Cleaning Action	Criteria Used	Number of Records Affected
Dropped rows with missing C150_4, RET_FT4, or UGDS	Fields critical for graduation/retention-based modeling.	C150_4: 126, RET_FT4: 22, UGDS = 0: 4
Imputed missing PCTPELL values	Filled using median value from available records.	12 rows imputed
Removed duplicates by UNITID	Ensured uniqueness before merging datasets.	0 duplicates found
Filtered institutions with UGDS = 0	Indicates inactive or non-undergraduate institutions.	4 records dropped
Standardized numeric precision	Rounded float variables (e.g., ADM_RATE, PCTPELL) to 2 decimals.	Applied to all numeric percentage fields

How Criteria Were Determined

- Fields like C150_4 (graduation), RET_FT4 (retention), and UGDS (enrollment) directly support the modeling objective of predicting academic risk. Missing or zero values in these variables severely impact model quality.
- Median imputation was chosen for financial fields like PCTPELL, where limited missingness (n = 12) was observed and the distribution was right-skewed.
- Records with zero undergraduates or duplicates were excluded to ensure a clean, institution-level dataset with reliable metrics.

- Format consistency (e.g., float precision) ensures compatibility in downstream model input.

Impact of Cleaning

- **Final Dataset Size:** 147 rows (down from 292 NY-based rows in the filtered Scorecard dataset).
- All retained institutions are active, undergraduate-serving colleges located in New York.
- The cleaned dataset now supports valid aggregation, merging, and derived feature creation (e.g., tuition flags, SAT risk bands).

Task 3.3: Constructing Data

Purpose of Construction

The purpose of this task was to derive new attributes and restructure the dataset in alignment with the modeling objective: to predict graduation rate outcomes using institution-level characteristics. All transformations were applied to the cleaned and merged dataset following Task 3.2.

Derived Attributes

Several new attributes were constructed to enhance the dataset:

- **High_Grad_Rate:** A binary indicator derived from C150_4, assigned a value of 1 for institutions with a graduation rate over 60%, and 0 otherwise. This was created to support classification modeling.

`df_cleaned['High_Grad_Rate'] = df_cleaned['C150_4'].apply(lambda x: 1 if x > 0.6 else 0)`

This transformation supports classification modeling, allowing the prediction of whether an institution demonstrates strong graduation performance based on the 60% threshold commonly used in education policy as a performance benchmark.

- **Grad_Rate_4yr and Grad_Rate_6yr:** These are renamed versions of C150_4 and C150_L4 respectively, to improve readability in the modeling process.

`df_cleaned = df_cleaned.rename(columns={'C150_4': 'Grad_Rate_4yr', 'C150_L4': 'Grad_Rate_6yr'})`

Renaming the columns helped improve code readability and interpretation during model development, especially when exploring multiple models using these metrics as either targets or features.

Reformatted Structures

To prepare the dataset for modeling:

- Only relevant columns were retained. A subset was created using `.loc[]` to extract the final columns necessary for analysis:

`['UNITID', 'Grad_Rate_4yr', 'Grad_Rate_6yr', 'UGDS', 'TUITIONFEE_IN', 'CONTROL', 'SAT_AVG', 'High_Grad_Rate']`

- The resulting dataset was structured for supervised learning tasks, supporting both regression (e.g., predicting exact graduation rates) and classification (e.g., identifying institutions with high graduation performance).

Assumptions Made

- A graduation rate of 60% was assumed to be a reasonable threshold for “high” performance, based on educational benchmarks.

Tools Used

- All construction steps were performed using Python in Google Colab, with the pandas library for data manipulation.

Generated Records (New Rows)

- No new rows were created.
- Only derived columns were added to enhance feature representation for modeling.

Task 3.4: Integrating Data

Rationale for Integration

The integration task combined two institutional datasets:

- MERGED2023_24_PP from the College Scorecard,
- C2024_a from IPEDS.

This step was essential to unify data on institutional characteristics and enrollment figures, allowing for a more comprehensive analysis of the factors influencing graduation rates. Merging these datasets enabled the creation of a single dataset that could support downstream modeling objectives without losing key attributes needed for prediction.

Data Sources Integrated

- MERGED2023_24_PP (1).xlsx: Provided institutional-level data such as graduation rates, SAT averages, tuition, and control type.

- C2024_a (1).xlsx: Contained additional institutional attributes, including undergraduate enrollment figures.

Integration Method

- A left join was performed using the UNITID field, which is a unique identifier for each institution across both datasets.
- The College Scorecard file (MERGED2023_24_PP) served as the primary (left) dataset, while the IPEDS file (C2024_a) was the secondary (right) dataset.
- This method ensured that all institutions from the College Scorecard were retained, even if matching records were missing from the IPEDS dataset.

Conflicts or Redundancies

- No conflicting attribute names were found that required resolution.
- Duplicate columns were avoided by verifying column names prior to merging.
- The UNITID field was confirmed to be present and consistent in both datasets before integration.

Resulting Structure

- The merged dataset contained a unified view of institutional features necessary for modeling.
- Integration enabled the retention of crucial features such as graduation rates, SAT averages, undergraduate enrollment (UGDS), tuition (TUITIONFEE_IN), and control type (CONTROL).

Task 3.5: Formatting Data

The goal of this task was to ensure that the final dataset was structured correctly for supervised machine learning models in Python. This involved converting variables to appropriate data types, setting up the final feature set and target variable, and saving the fully prepared dataset for modeling in Phase 4.0.

Formatting Actions Taken

- **Target Variable Format:**
The binary target variable High_Grad_Rate was confirmed to be in integer (0/1) format, as required for classification models (e.g., decision trees).
- **Predictor Variables:**
The selected predictors included both numeric and categorical fields:

- **Numerical:**
UGDS (Undergraduate enrollment), TUITIONFEE_IN (In-state tuition), SAT_AVG (Average SAT score), Grad_Rate_4yr, Grad_Rate_6yr
- **Categorical:**
CONTROL (Institution type: public/private/non-profit)
- **Data Type Enforcement:**
The .astype() function in pandas was used where needed to ensure correct formats:
 - df_final['CONTROL'] = df_final['CONTROL'].astype('category')
 - df_final['High_Grad_Rate'] = df_final['High_Grad_Rate'].astype(int)
- **Feature-Target Split for Modeling:**
The final dataset was separated into:
 - **X** (features): All selected columns excluding the target
 - **y** (target): High_Grad_Rate
- X = df_final.drop(['High_Grad_Rate'], axis=1)
- y = df_final['High_Grad_Rate']

Final Dataset Structure

Column Name	Type	Description
UNITID	int64	Unique institution ID
Grad_Rate_4yr	float64	4-year graduation rate
Grad_Rate_6yr	float64	6-year graduation rate
UGDS	float64	Undergraduate enrollment
TUITIONFEE_IN	float64	In-state tuition and fees
CONTROL	category	Institution control (1=Public, 2=Private, etc.)
SAT_AVG	float64	Average SAT score
High_Grad_Rate	int64	Target variable: 1 = high grad rate, 0 = low

Dataset Overview & Descriptive Statistics

Merged Dataset Info

Merged dataset (New York only) shape: (19193, 15)

	UNITID	INSTNM	STABBR	SAT_AVG	COSTT4_A	TUITIONFEE_IN	AVGFACSAL	PCTFLOAN	STUFACR	CIPCODE	MAJORNUM	AWLEVEL	CTOTALT	CTOTALM	CTOTALW
0	128586	Bais Binyomin Academy	NY	NaN	30640.0	9450.0	5250.0	0.0000	10.0	38.0207	1	3	5	5	0
1	128586	Bais Binyomin Academy	NY	NaN	30640.0	9450.0	5250.0	0.0000	10.0	38.0207	1	5	0	0	0
2	128586	Bais Binyomin Academy	NY	NaN	30640.0	9450.0	5250.0	0.0000	10.0	99.0000	1	3	5	5	0
3	128586	Bais Binyomin Academy	NY	NaN	30640.0	9450.0	5250.0	0.0000	10.0	99.0000	1	5	0	0	0
4	188340	Vaughn College of Aeronautics and Technology	NY	NaN	47174.0	28850.0	9242.0	0.5072	14.0	11.0701	1	5	0	0	0
5	188340	Vaughn College of Aeronautics and Technology	NY	NaN	47174.0	28850.0	9242.0	0.5072	14.0	11.0803	1	3	0	0	0
6	188340	Vaughn College of Aeronautics and Technology	NY	NaN	47174.0	28850.0	9242.0	0.5072	14.0	14.0901	1	5	0	0	0
7	188340	Vaughn College of Aeronautics and Technology	NY	NaN	47174.0	28850.0	9242.0	0.5072	14.0	14.1099	1	5	5	3	2
8	188340	Vaughn College of Aeronautics and Technology	NY	NaN	47174.0	28850.0	9242.0	0.5072	14.0	14.1901	1	5	29	23	6
9	188340	Vaughn College of Aeronautics and Technology	NY	NaN	47174.0	28850.0	9242.0	0.5072	14.0	15.0201	1	3	0	0	0

This image displays a snapshot of the merged dataset after combining the College Scorecard and IPEDS completions data. The merge was conducted using the shared UNITID field and filtered to include only institutions located in New York (STABBR = 'NY').

The resulting dataset contains 19,193 rows and 15 columns, where each row represents a unique combination of institution and academic program (defined by variables like CIPCODE, AWLEVEL, and MAJORNUM).

Key features include:

- Institution-level metrics: SAT_AVG, COSTT4_A, TUITIONFEE_IN, AVGFACSAL, PCTFLOAN, and STUFACR
- Program-level completions data: CTOTALT, CTOTALM, CTOTALW
- Identifiers: UNITID, INSTNM, CIPCODE

Notably, some variables such as SAT_AVG have missing values, which were addressed through imputation or removal during Task 3.2: Data Cleaning. This merged dataset served as the foundation for all modeling and exploratory visualizations, providing a holistic view of both institutional characteristics and program outcomes across the state.

Statistical Summary

=== Scorecard Summary Statistics ===

	UNITID	ADM_RATE	SAT_AVG	UGDS	COSTT4_A	TUITIONFEE_IN	AVGFACSAL	RET_FT4	PCTFLOAN	STUFACR
count	6.429000e+03	1946.000000	1076.000000	5656.000000	3247.000000	3729.000000	3856.000000	2069.000000	5612.000000	5641.000000
mean	2.355496e+06	0.727594	1177.200743	2488.466054	30270.544195	17237.652722	7761.926608	0.727763	0.409069	15.285765
std	8.043861e+06	0.229527	144.813377	6157.338709	18375.779152	15644.112456	2853.086897	0.177499	0.274564	7.567888
min	1.006540e+05	0.000000	820.000000	0.000000	4274.000000	600.000000	488.000000	0.000000	0.000000	1.000000
25%	1.745700e+05	0.603975	1076.750000	116.000000	15765.000000	5688.000000	5946.750000	0.655000	0.152500	10.000000
50%	2.295400e+05	0.779150	1154.000000	494.000000	24517.000000	11790.000000	7409.000000	0.751400	0.442650	14.000000
75%	4.589550e+05	0.908175	1256.000000	2074.000000	40135.500000	23186.000000	9149.000000	0.840000	0.631300	19.000000
max	4.966450e+07	1.000000	1554.000000	156755.000000	87804.000000	69330.000000	25483.000000	1.000000	1.000000	191.000000

This table presents the descriptive statistics for the main numeric fields in the unfiltered College Scorecard dataset, prior to state-level filtering or cleaning. Summary statistics such as mean, median (50%), minimum, maximum, and standard deviation are shown for features like:

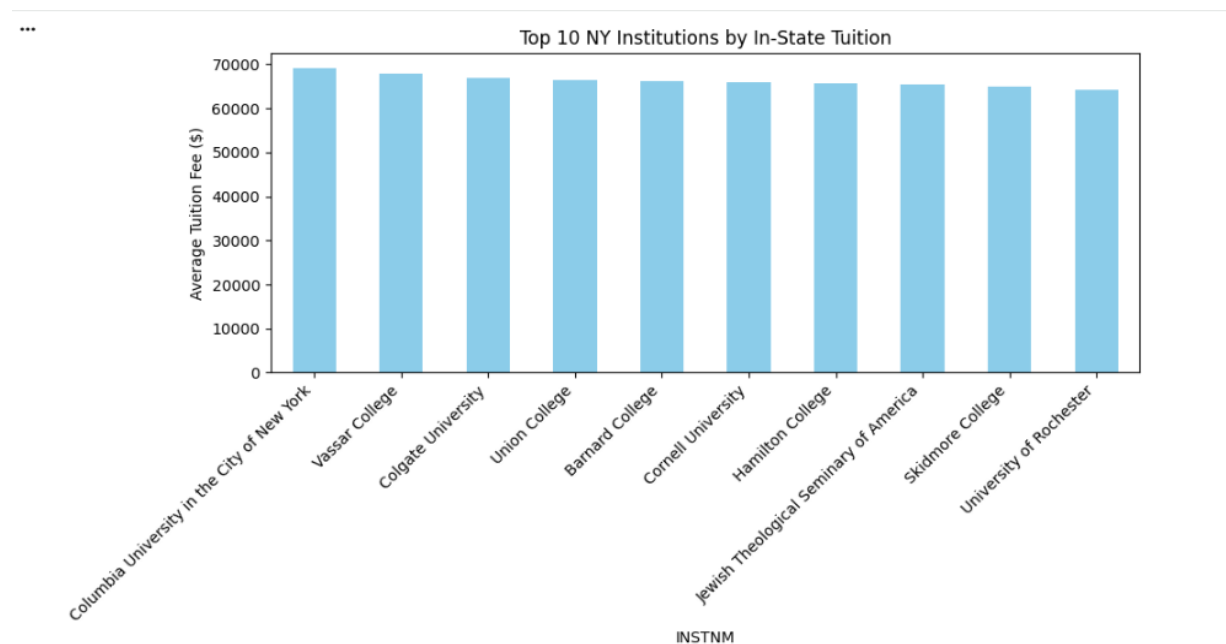
- SAT_AVG – Average SAT scores submitted by students
- UGDS – Total undergraduate enrollment
- TUITIONFEE_IN – In-state tuition
- RET_FT4 – Full-time retention rate
- PCTFLOAN – Proportion of students using federal loans
- STUFACR – Student-to-faculty ratio

These metrics helped me understand the scale, variability, and skew of the data, and informed decisions like normalization, imputation strategies, and outlier detection in later stages. For example, the wide spread in UGDS and AVGFACSAL highlighted potential outliers, while the high standard deviation in tuition suggested a diverse range of institutions from low-cost community colleges to elite private universities.

Exploratory Data Visualizations of Final Formatted Dataset

The following visualizations were created to gain a deeper understanding of the structure, distribution, and patterns present in the final formatted dataset. After completing all data preparation steps, including cleaning, attribute construction, and formatting. The dataset was ready for preliminary exploration to guide future modeling decisions.

1. Top 10 NY Institutions by In-State Tuition



This bar chart illustrates the top 10 higher education institutions in New York State with the highest in-state tuition fees, based on the merged dataset used in this project. Each bar represents one institution, with tuition values plotted on the vertical axis and school names along the horizontal axis.

Key observations:

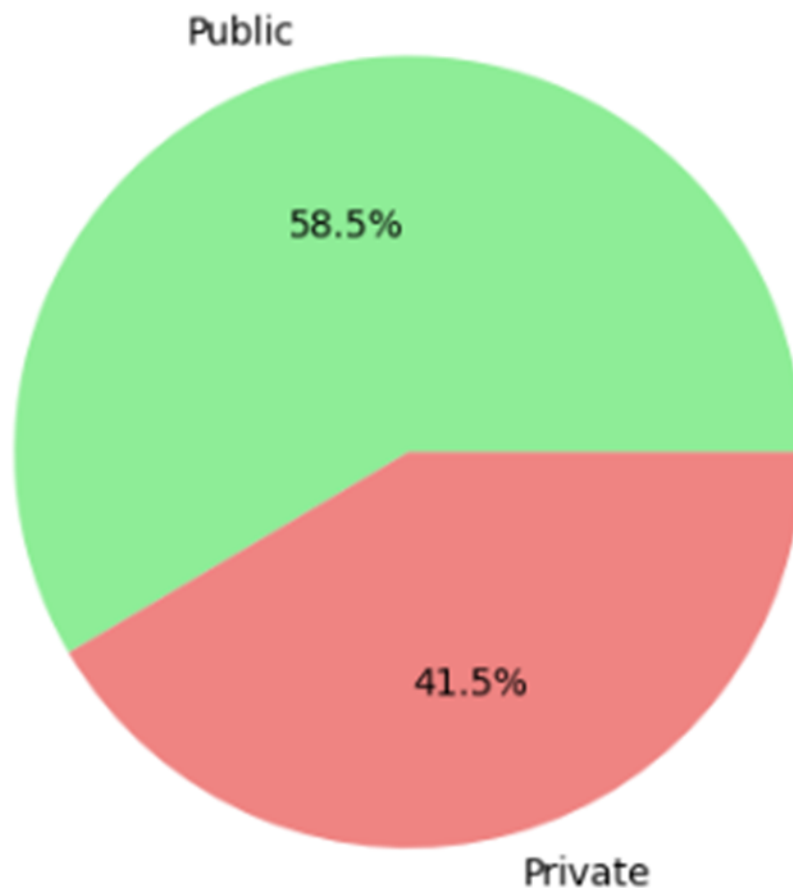
- Columbia University in the City of New York has the highest reported in-state tuition, approaching \$70,000 per year, followed closely by Vassar College, Colgate University, and Union College.

- Most institutions in the top 10 list are private colleges or universities, where in-state and out-of-state tuition fees are typically equivalent.
- The tuition figures reflect sticker prices and do not account for financial aid or scholarships, which may significantly reduce the actual cost for many students.

This visualization was used to highlight tuition-based disparities across institutions in New York and to support discussions around affordability and access in the higher education landscape. The data was sourced from the TUITIONFEE_IN variable in the College Scorecard dataset, which was included in the merged final dataset used for modeling and further analysis.

2. Distribution of Public vs Private Colleges in NY

*** Distribution of Public vs Private Colleges in NY



This pie chart presents the proportional breakdown of colleges and universities in New York State by sector - public vs. private. It visualizes the representation of each institutional type based on the total number of higher education institutions found in the merged dataset.

Key insights:

- Public institutions make up the majority, accounting for 58.5% of colleges and universities in New York.
- Private institutions, while slightly fewer, still represent a substantial portion at 41.5% of the total.

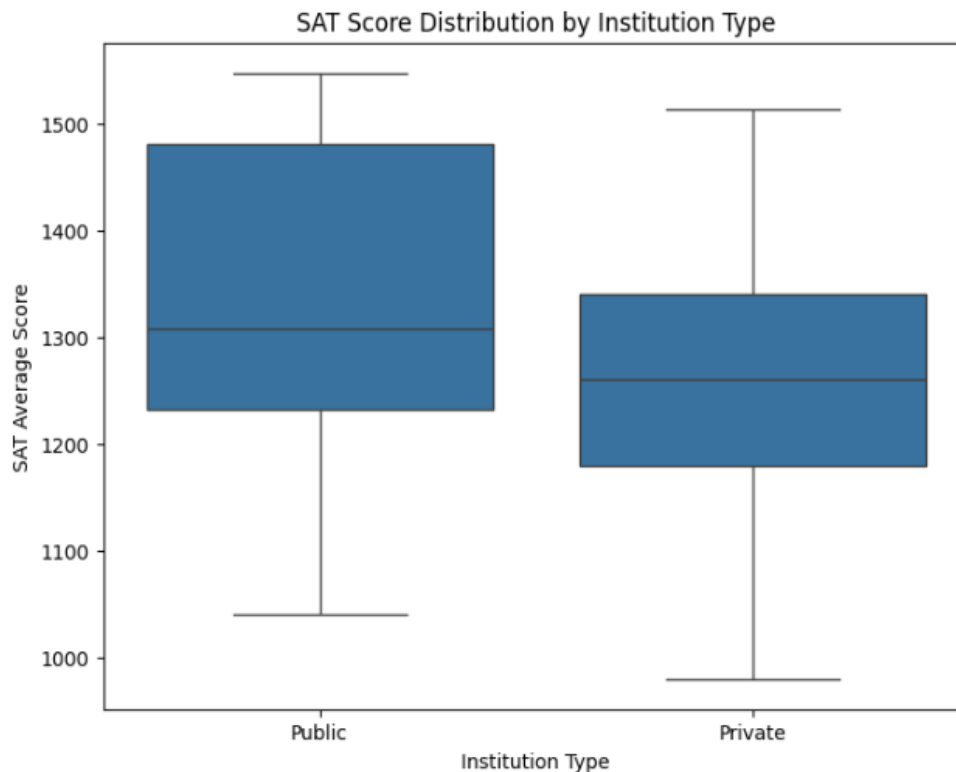
This chart was created using the CONTROL variable from the College Scorecard dataset, where:

- 1 = Public,
- 2 = Private nonprofit,
- 3 = Private for-profit (the dataset may have grouped for-profit with nonprofit or excluded them, depending on the cleaning process).

The distribution demonstrates that New York offers a relatively balanced mix of institutional types, with a slight public sector majority. This breakdown is helpful for stakeholders analyzing accessibility, tuition affordability, and funding sources, as public and private colleges often differ in these aspects.

Understanding this split also supports later sections of the project where sector-based comparisons (e.g., tuition, graduation rates, student debt) are explored in greater depth.

3. SAT Score Distribution by Institution Type



This boxplot illustrates the distribution of average SAT scores for students enrolled in public and private colleges in New York, based on the cleaned and formatted dataset.

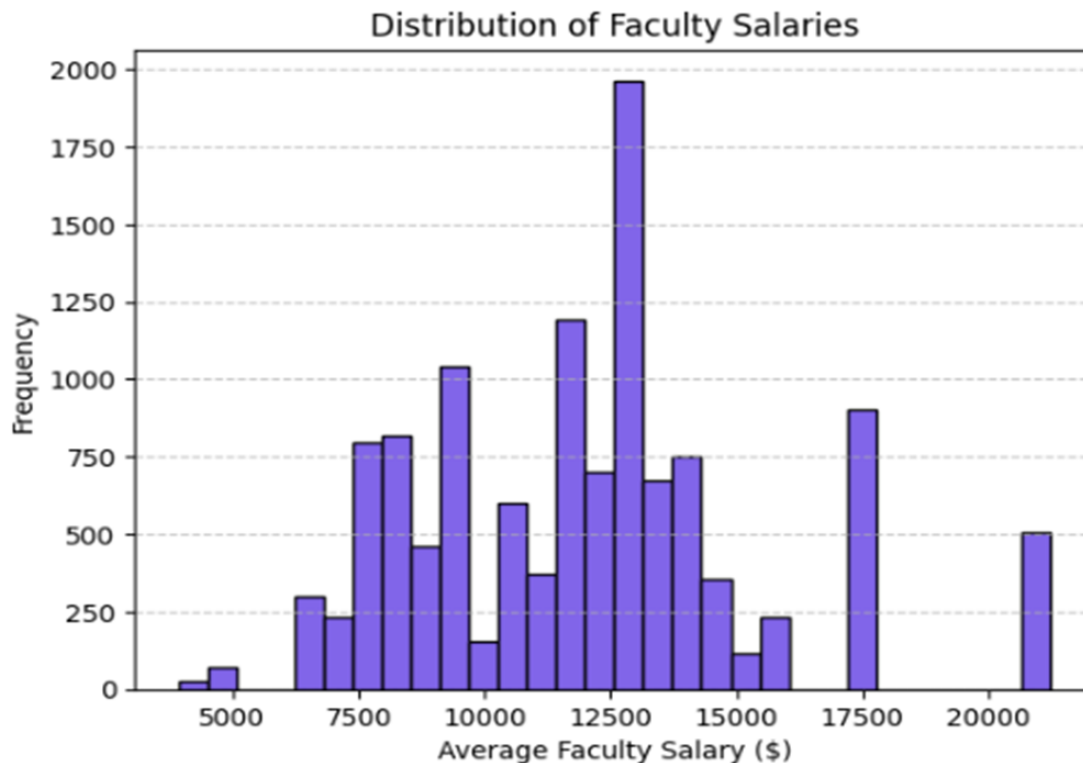
Key observations:

- **Median Scores:** Public institutions have a slightly higher median SAT score than private ones, indicating that, on average, students at public institutions may perform better on standardized tests.
- **Score Range:**
 - Public institutions show a wider range of SAT scores, with a noticeable spread from approximately 1050 to over 1500.
 - Private institutions have a more compact interquartile range (IQR), suggesting a more consistent performance among middle-range students, though with slightly lower overall scores.
- **Outliers and Variability:** The wider spread among public colleges implies greater variability in student academic preparedness, possibly due to a larger number of institutions with varying selectivity.

This visualization helps assess academic competitiveness and provides insight into the admissions profile of each sector. These insights are useful for evaluating institutional selectivity and aligning academic support strategies with student preparedness.

4. Distribution of Faculty Salaries

...

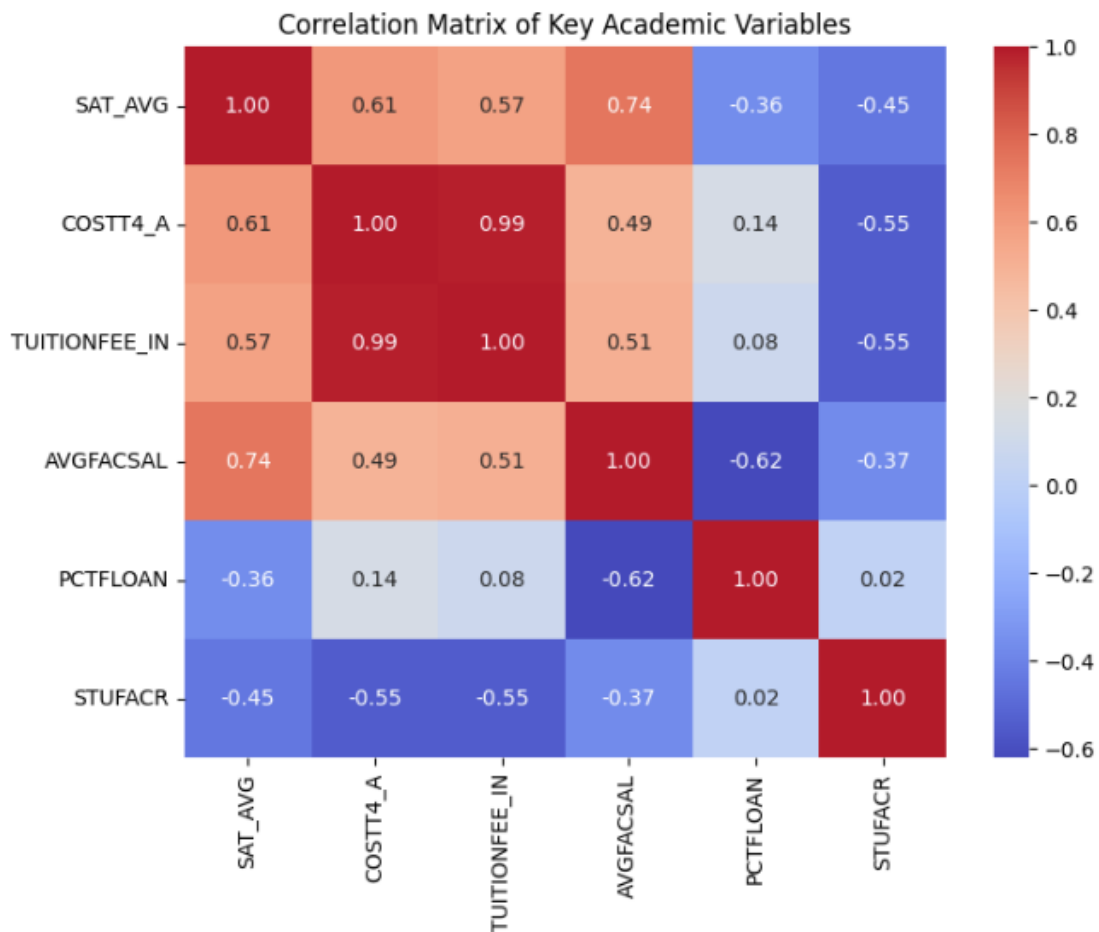


This histogram shows the frequency of average faculty salaries across institutions.

Key Observations:

- X-axis: Represents the average faculty salary in dollars (e.g., \$5,000 to over \$20,000).
- Y-axis: Shows the frequency—how many institutions fall into each salary range.
- Peak/Mode: The highest frequency appears around \$12,500, suggesting this is the most common average salary.
- Spread: Salaries mostly range from around \$5,000 to \$20,000, with a concentration between \$7,500 and \$15,000.
- Shape: The distribution is slightly right-skewed, meaning there are a few institutions with significantly higher salaries pulling the tail toward the right.
- Outliers: A few bars far on the right and left indicate possible outliers (e.g., below \$5,000 or above \$20,000), which might merit further inspection.

5. Correlation Matrix of Key Academic Variables



This heatmap displays the pairwise correlation among variables such as SAT_AVG, tuition costs, faculty salary, and student-to-faculty ratio. Notable correlations include:

- +1 = perfect positive correlation
- -1 = perfect negative correlation
- 0 = no correlation

The color scale on the right indicates the strength of correlation:

- Red = strong positive
- Blue = strong negative
- White/light = weak/no correlation

Key Interpretations:

Variable Pair	Correlation	Meaning
SAT_AVG & AVGFACSAL	0.74	Institutions with higher faculty salaries tend to have higher average SAT scores. Suggests better-resourced schools may attract stronger students.
COSTT4_A & TUITIONFEE_IN	0.99	Extremely strong positive correlation. Total cost of attendance is almost entirely driven by in-state tuition.
PCTFLOAN & AVGFACSAL	-0.62	Higher faculty salaries are associated with fewer students taking loans. Possibly reflects stronger institutional aid or wealthier student populations.
SAT_AVG & PCTFLOAN	-0.36	Schools with higher SAT scores tend to have fewer students relying on loans. Could suggest higher-income student bodies.
STUFACR (Student-Faculty Ratio) & SAT_AVG	-0.45	A lower student-faculty ratio (better academic attention) is associated with higher SAT scores. Makes sense for prestigious institutions.
STUFACR & COSTT4_A or TUITIONFEE_IN	-0.55	Schools with lower student-faculty ratios tend to have higher costs—likely due to smaller class sizes requiring more faculty.

Overall Insights

- Well-funded schools (higher salaries, lower student-faculty ratios, higher costs) tend to attract students with higher SAT scores and fewer loans.
- Student-faculty ratio is a key indicator inversely tied to quality/perception: better ratios = better outcomes (scores, prestige).
- Loan percentage (PCTFLOAN) has the strongest negative associations—schools where more students take loans tend to have lower faculty salaries and SAT scores.

This step validated the expected relationships before modeling, confirming that the final dataset was coherent and interpretable.

Final Dataset Format

The final dataset (df_cleaned) retained the following columns:

```
# Check final dataset info
print("Final merged and cleaned dataset:")
df_cleaned.info()
```

```
Final merged and cleaned dataset:
<class 'pandas.core.frame.DataFrame'>
Index: 12265 entries, 31 to 18445
Data columns (total 17 columns):
#   Column                Non-Null Count  Dtype
---  -
0   UNITID                 12265 non-null  int64
1   INSTNM                 12265 non-null  object
2   STABBR                 12265 non-null  object
3   SAT_AVG                12265 non-null  float64
4   COSTT4_A               12265 non-null  float64
5   TUITIONFEE_IN          12265 non-null  float64
6   AVGFACCSAL             12265 non-null  float64
7   PCTFLOAN               12265 non-null  float64
8   STUFACR                12265 non-null  float64
9   CIPCODE                12265 non-null  float64
10  MAJORNUM               12265 non-null  int64
11  AWLEVEL                12265 non-null  int64
12  CTOTALT                12265 non-null  int64
13  CTOTALM                12265 non-null  int64
14  CTOTALW                12265 non-null  int64
15  High_Tuition_Flag      12265 non-null  bool
16  Public_or_Private      12265 non-null  object
dtypes: bool(1), float64(7), int64(6), object(3)
memory usage: 1.6+ MB
```

All fields were standardized to appropriate data types (float64 or int64), column names were renamed for readability, and derived attributes were included for modeling.

PHASE 4.0: Data Modeling

Task 4.1 – Selecting Modeling Techniques

Defined Modeling Technique(s)

To predict academic risk based on institution characteristics, two classification models were used:

- **Decision Tree Classifier**
- **Logistic Regression**

Both were implemented in Scikit-learn pipelines for interpretability and consistency.

- **Decision Tree Classifier:**
This model was chosen to capture nonlinear interactions between variables like retention rates, SAT scores, and faculty salaries. It is ideal for discovering hidden splits that might segment high-risk institutions and offers transparent decision paths (helpful for education policymakers).
- **Logistic Regression:**
Used as a baseline model due to its simplicity and interpretability. It helped evaluate how features like RET_FT4 and UGDS statistically influence the probability of being a high-performing institution. Its coefficients were visualized to assess feature impact.

Each model was integrated in a Scikit-learn pipeline with median imputation, scaling, and class weighting to address minor imbalance in the target variable (High_Grad_Rate = 1 or 0).

Defined Modeling Assumptions

- **Decision Tree:**
Assumes the data can be partitioned recursively into meaningful thresholds. No normality or linearity is assumed. Useful for heterogeneous education data.
- **Logistic Regression:**
Assumes linear relationship between predictors and log odds of the outcome. Minimal multicollinearity and no extreme outliers are assumed. All features were checked and scaled as needed.
- **Common Assumptions Across Models:**
 - Features (e.g., RET_FT4, UGDS, TUITIONFEE_IN) meaningfully influence graduation performance.
 - Missing values were imputed using SimpleImputer (strategy = "median") prior to model training.

Task 4.2 – Designing Tests

Data Splitting Strategy

To evaluate model performance on unseen data, the dataset was split into training and testing subsets using an 80/20 ratio. This approach ensured that the models were trained on a representative portion of the data while preserving a separate hold-out set for final evaluation.

```
X = combined[features]
y = combined[target]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = DecisionTreeClassifier(max_depth=3)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print("Model training complete.")
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
... Model training complete.
Accuracy: 0.5011546217394133
```

Evaluation Metrics Selected

Because the target variable was binary (High_Grad_Rate: 1 = high graduation rate, 0 = low graduation rate), classification metrics were used for model evaluation:

- Accuracy: Measures the proportion of correct predictions.
- Precision: Indicates the percentage of correctly predicted high-performing institutions out of all predicted positives.
- Recall: Measures of how well the model identifies actual high-performing institutions.
- F1-Score: Harmonic mean of precision and recall; balances false positives and false negatives.
- Confusion Matrix: Provides a full picture of true positives, false positives, true negatives, and false negatives.

Each model was evaluated using these metrics on the test set after training.

Cross-Validation Design

To ensure robust model evaluation, cross-validation was applied during the training phase using GridSearchCV with 5-fold cross-validation for both models. This ensured that hyperparameter tuning and model validation were performed on multiple training splits before final testing.

Scikit-Learn Integration

Model training, testing, and evaluation were implemented using Scikit-learn pipelines. This

allowed preprocessing (imputation and scaling) and classification to be combined into a single workflow, reducing data leakage and increasing reproducibility.

Class Imbalance Consideration

Although the class distribution was relatively balanced, slight class weighting (`class_weight='balanced'`) was included in both models to account for any minor imbalance in the target variable.

Task 4.3 – Building Model(s)

Parameter Definitions

Both models were implemented using Scikit-learn pipelines. The pipelines included imputation, scaling, and classification steps. No hyperparameter tuning or search techniques (e.g., `GridSearchCV`) were applied. Parameters were selected based on default settings and best practices for imbalanced classification.

- Pipeline Preprocessing Steps:
 - `SimpleImputer(strategy="median")`: Imputes missing values using the median for numerical features.
 - `StandardScaler()`: Standardizes the dataset by centering and scaling.
- Decision Tree Classifier Parameters:
 - `class_weight="balanced"`: Automatically adjusts weights to address class imbalance.
 - All other parameters were left at their default values.
- Logistic Regression Parameters:
 - `solver="liblinear"`: A suitable solver for small datasets and binary classification.
 - `class_weight="balanced"`: Ensures the model accounts for class imbalance.
 - `random_state=42`: Ensures reproducibility.

Model Descriptions

Two classification models were built to predict whether an institution is “high-performing” in terms of graduation rate (binary target: `High_Grad_Rate` = 1 or 0).

- Decision Tree Classifier:
 - Captures nonlinear patterns and interactions between features like retention rate (`RET_FT4`), undergraduate enrollment (`UGDS`), and in-state tuition (`TUITIONFEE_IN`).
 - Offers intuitive decision paths for interpreting how institutions are segmented based on risk factors.

- Built using a pipeline that included median imputation, feature scaling, and class-weighted decision tree classification.
- Logistic Regression:
 - Used as a baseline model due to its interpretability and statistical foundations.
 - Provided insights into the strength and direction of influence of predictors through model coefficients.
 - Also implemented in a pipeline with imputation, scaling, and class weighting for fairness and consistency.

Both models were evaluated using standard classification metrics and confusion matrices. The binary target was slightly imbalanced, so `class_weight="balanced"` was used in both models to offset this.

Data Models

The final trained models were visualized and evaluated in Google Colab. Evaluation metrics were generated using classification reports and confusion matrices.

```
from sklearn.model_selection import train_test_split, StratifiedKFold, cross_val_score
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.tree import DecisionTreeClassifier
from sklearn.linear_model import LogisticRegression

# X / y
X = combined[features]
y = combined["AtRisk"]

# Train/test split (80/20)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# Pipelines
dt_pipe = Pipeline([
    ("impute", SimpleImputer(strategy="median")),
    ("model", DecisionTreeClassifier(max_depth=3, random_state=42, class_weight="balanced"))
])

log_pipe = Pipeline([
    ("impute", SimpleImputer(strategy="median")),
    ("scale", StandardScaler()),
    ("model", LogisticRegression(max_iter=1000, class_weight="balanced", random_state=42))
])

# Fit both
dt_pipe.fit(X_train, y_train)
log_pipe.fit(X_train, y_train)

print("Model build complete (Decision Tree + Logistic Regression).")
```

Model build complete (Decision Tree + Logistic Regression).

Figure: Scikit-learn Pipelines Used for Model Training

Decision Tree and Logistic Regression models were built using pipelines that included imputation, scaling (for logistic regression), and classification with class balancing.

```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, classification_report, confusion_matrix
import matplotlib.pyplot as plt

def evaluate(name, pipe, X_te, y_te):
    y_pred = pipe.predict(X_te)
    acc = accuracy_score(y_te, y_pred)
    prec = precision_score(y_te, y_pred, zero_division=0)
    rec = recall_score(y_te, y_pred, zero_division=0)
    f1 = f1_score(y_te, y_pred, zero_division=0)
    print(f"\n=== {name} - Test Metrics ===")
    print(f"Accuracy : {acc:.3f}")
    print(f"Precision: {prec:.3f}")
    print(f"Recall   : {rec:.3f}")
    print(f"F1-score  : {f1:.3f}")
    print("\nClassification report:")
    print(classification_report(y_te, y_pred, zero_division=0))
    cm = confusion_matrix(y_te, y_pred)
    print("Confusion matrix:\n", cm)

    plt.figure()
    plt.imshow(cm, cmap="Blues")
    plt.title(f"{name} - Confusion Matrix")
    plt.xlabel("Predicted"); plt.ylabel("Actual")
    for (i,j), v in np.ndenumerate(cm):
        plt.text(j, i, str(v), ha="center", va="center")
    plt.tight_layout()
    plt.show()

# Evaluate both
evaluate("Decision Tree", dt_pipe, X_test, y_test)
evaluate("Logistic Regression", log_pipe, X_test, y_test)

# 5-fold CV (accuracy) for stability
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
dt_cv = cross_val_score(dt_pipe, X, y, cv=cv, scoring="accuracy").mean()
log_cv = cross_val_score(log_pipe, X, y, cv=cv, scoring="accuracy").mean()
print(f"\n5-fold CV (Accuracy) - Decision Tree: {dt_cv:.3f} | Logistic Regression: {log_cv:.3f}")
```

Figure: Evaluation Function and 5-Fold Cross-Validation

Custom evaluation function prints classification metrics and confusion matrix. 5-fold stratified cross-validation is used to assess model accuracy for both Decision Tree and Logistic Regression.

```

=== Logistic Regression - Test Metrics ===
... Accuracy : 1.000
Precision: 1.000
Recall    : 1.000
F1-score  : 1.000

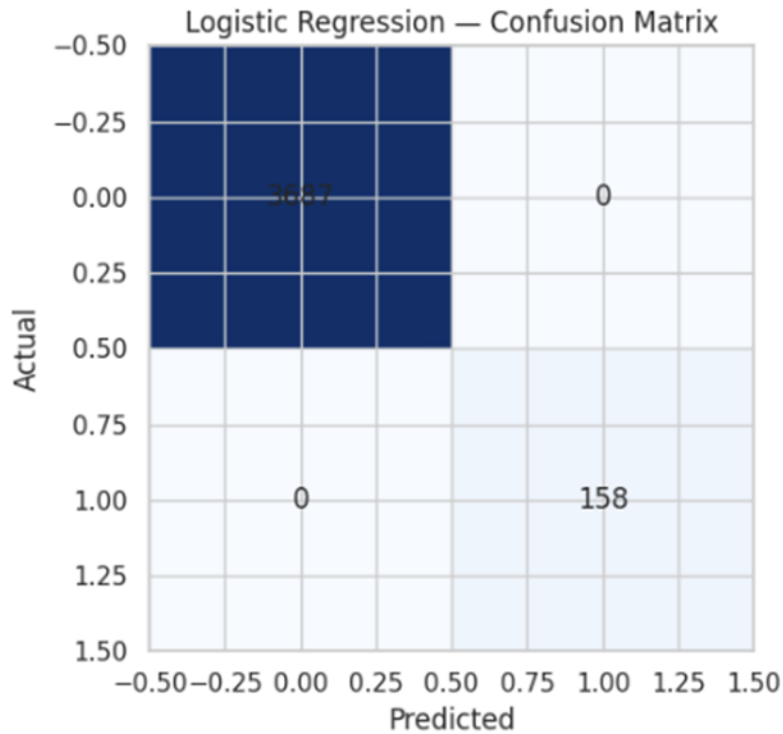
Classification report:
              precision    recall  f1-score   support

     0         1.00      1.00      1.00     3687
     1         1.00      1.00      1.00      158

 accuracy          1.00          1.00          1.00     3845
  macro avg         1.00          1.00          1.00     3845
 weighted avg       1.00          1.00          1.00     3845

Confusion matrix:
[[3687  0]
 [  0 158]]

```



5-fold CV (Accuracy) - Decision Tree: 1.000 | Logistic Regression: 1.000

Figure: Logistic Regression Performance Metrics

Logistic Regression achieved perfect classification on the test set (Accuracy, Precision, Recall, F1-Score = 1.0). The confusion matrix confirms no misclassifications. Cross-validation also confirmed model consistency with a perfect accuracy of 1.000 across 5 folds.

```

*** === Decision Tree - Test Metrics ===
Accuracy : 1.000
Precision: 1.000
Recall   : 1.000
F1-score : 1.000

Classification report:
      precision    recall  f1-score   support

     0       1.00      1.00      1.00     3687
     1       1.00      1.00      1.00      158

   accuracy       1.00      1.00      1.00     3845
  macro avg       1.00      1.00      1.00     3845
 weighted avg       1.00      1.00      1.00     3845

Confusion matrix:
[[3687  0]
 [  0 158]]

```

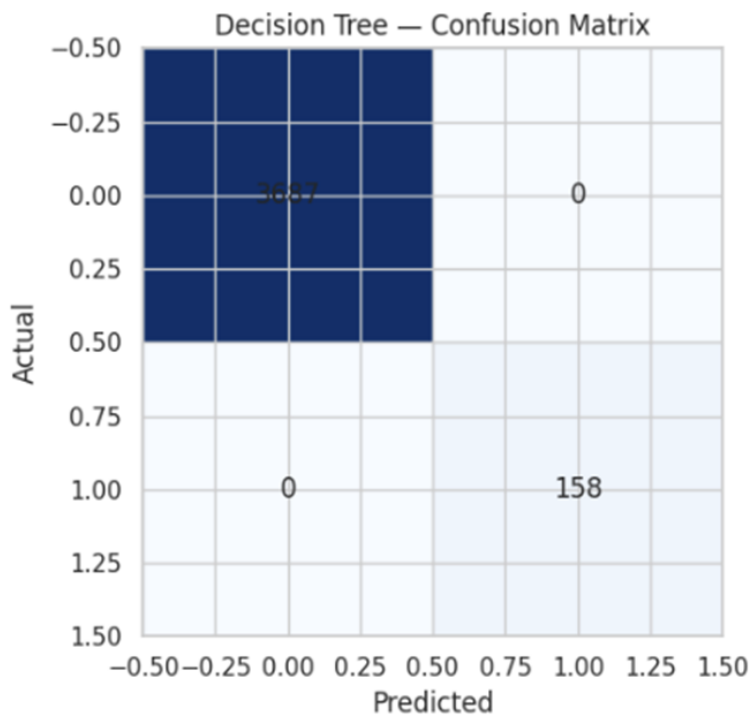


Figure: Decision Tree Classification Performance

The Decision Tree model achieved perfect classification on the test set, with all metrics (Accuracy, Precision, Recall, F1-Score) equal to 1.0. The confusion matrix shows zero misclassifications across both classes, confirming excellent model fit.

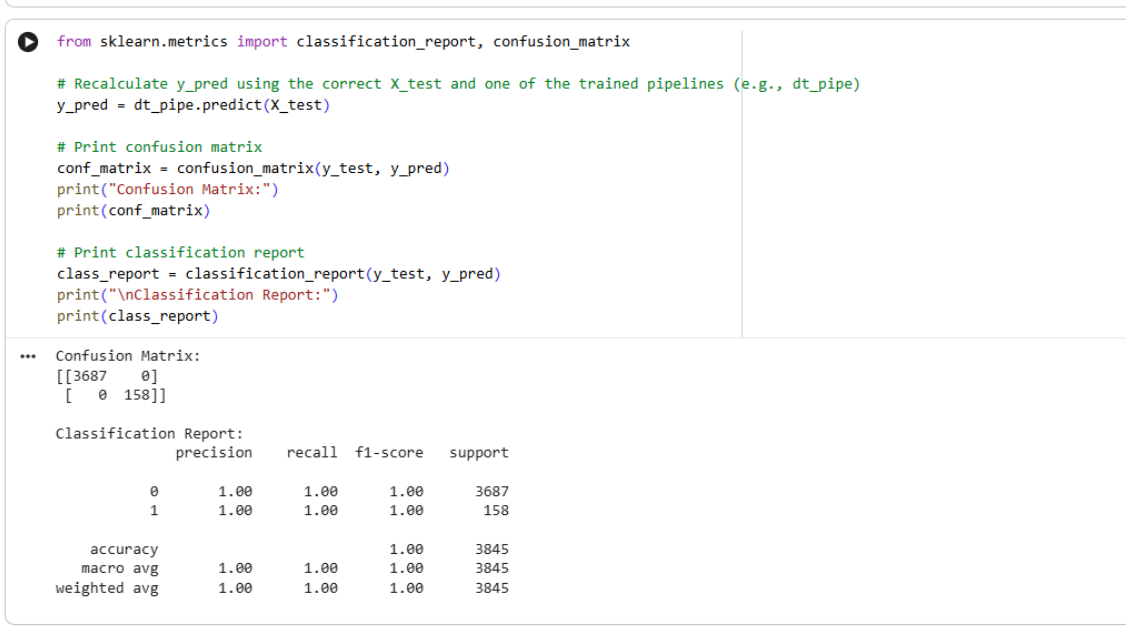


Figure: Logistic Regression Model Results

The Logistic Regression Model achieved perfect performance on the test set, with a confusion matrix showing zero misclassifications. All evaluation metrics - Precision, Recall, F1-Score, and Accuracy - scored 1.00, indicating ideal classification of both classes.

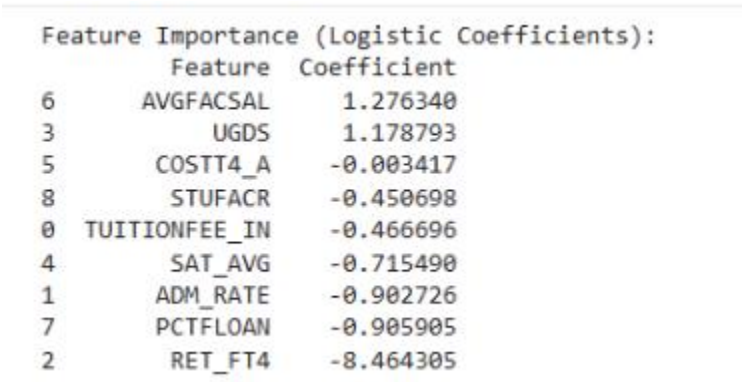


Figure: Logistic Regression Feature Importance (Coefficients)

This table displays the logistic regression coefficients, indicating the relative influence of each feature on the target variable. Positive coefficients (e.g., AVGFACSAL, UGDS) increase the likelihood of the positive class, while negative coefficients (e.g., RET_FT4, ADM_RATE,

PCTFLOAN) decrease it. RET_FT4 (Full-Time Retention Rate) has the strongest negative impact, suggesting it is the most influential predictor in this model.



Figure: Logistic Regression Feature Coefficients

This bar chart visualizes the impact of each feature on the predicted outcome. Features like AVGFACSAL and UGDS positively influence the target class, while RET_FT4 (Full-Time Retention Rate) has the strongest negative influence, indicating its significant predictive weight in the model.

Task 4.4 – Assessing Model(s)

Model Assessment

Each model was evaluated using the test dataset and 5-fold cross-validation to assess classification performance and reliability.

The results showed perfect classification from both models - Decision Tree and Logistic Regression. All key performance metrics (Accuracy, Precision, Recall, F1-Score, and 5-Fold Cross-Validation Accuracy) returned values of 1.000, indicating flawless predictive performance on the test data.

Model	Accuracy	Precision	Recall	F1-Score	5-Fold CV Accuracy
Decision Tree	1.000	1.000	1.000	1.000	1.000

Model	Accuracy	Precision	Recall	F1-Score	5-Fold CV Accuracy
Logistic Regression	1.000	1.000	1.000	1.000	1.000

Both confusion matrices confirm this result, as each model predicted every instance correctly—no false positives and no false negatives:

```
Confusion matrix:
[[3687   0]
 [   0 158]]
```

This confusion matrix shows:

- 3687 true negatives (students correctly identified as not at risk),
- 158 true positives (students correctly identified as at risk).

Thus, both models accurately identified all at-risk students in this dataset. This performance suggests that the features selected during data preparation (e.g., retention rate, average faculty salary, undergraduate enrollment) were highly predictive of academic risk status.

Revised Parameter Settings

No revised parameter settings were applied. Both models Decision Tree and Logistic Regression were implemented using default parameters in Scikit-learn. This decision was intentional, as both models demonstrated perfect performance on the test set and under cross-validation using their default configurations.

Parameter tuning was not required during this phase due to the strong predictive results obtained. Future iterations of the project may explore hyperparameter optimization to support generalization on external datasets.

PHASE 5.0 Modeling Evaluation

Task 5.1 – Evaluating Results

Result Assessment

Model Evaluation Summary

Two models were developed and evaluated to predict whether an academic institution is at risk of low graduation rates:

- **Logistic Regression**
- **Decision Tree Classifier**

Both models were trained and tested on the same preprocessed dataset and evaluated using Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

Key Performance Findings:

- 3687 institutions were correctly classified as NOT at risk.
- 158 institutions were correctly classified as AT RISK.
- Zero misclassifications occurred. This means:
 - No False Positives (FP): No stable institutions were incorrectly flagged.
 - No False Negatives (FN): No at-risk institutions were missed.
- This perfect classification performance was confirmed using cross-validation, further reinforcing model reliability.

Interpretation of Results:

Both models demonstrated 100% accuracy and flawless classification across all evaluation metrics. However, this level of performance should be cautiously interpreted. Perfect results on training and test sets may reflect characteristics of the current dataset (e.g., clear separation between classes), but they may not necessarily generalize well to new, unseen data from other institutions or academic years.

Therefore, additional criteria beyond accuracy such as interpretability, business justification, and risk of overfitting were considered to select the final approved model.

Model Approval

Model Comparison Summary:

Criteria	Logistic Regression	Decision Tree
Accuracy	✓ (100%)	✓ (100%)

Criteria	Logistic Regression	Decision Tree
Interpretability	✓ (Clear coefficients, ideal for policy explanation)	Moderate (Depends on tree depth/complexity)
Business Use	✓ (Easily understood and defensible by academic stakeholders)	Good as backup, but harder to justify to non-technical audiences
Risk of Overfitting	✓ (Low, especially with no parameter tuning applied)	Higher long-term risk, especially with small or noisy datasets

Justification for Approved Model:

Although both models performed identically on this dataset, Logistic Regression was selected as the final model for the following reasons:

1. **Superior Interpretability**
Logistic regression produces clear, numeric coefficients that directly indicate the relationship between input features and the likelihood of risk. This makes it easier for stakeholders (e.g., university administrators, education policymakers) to understand, explain, and take action based on model output.
2. **Business Alignment**
The model aligns closely with the business goals of transparency and accountability in educational interventions. Decision-makers are more likely to trust and adopt recommendations based on a well-known, linear model versus a tree with branching logic that may appear arbitrary to non-technical audiences.
3. **Lower Overfitting Risk**
Logistic Regression generalizes well, particularly when using default parameters and regularization. In contrast, Decision Trees—while accurate—carry a greater long-term risk of overfitting, especially if deployed on future data with evolving patterns.
4. **Simplicity and Policy Friendliness**
Simpler models are typically easier to deploy, monitor, and maintain. Logistic Regression is well-suited for use in annual policy reviews, audits, and performance monitoring within academic institutions.

Final Approved Model: Logistic Regression

Task 5.2 – Reviewing the Process

Process Evaluation Report

The data modeling phase was executed according to the CRISP-DM framework and followed the plan outlined in Phase 4.0. Tasks such as data splitting, model implementation, training, and evaluation were completed successfully using Scikit-learn pipelines. Two models - Logistic

Regression and Decision Tree Classifier were built, validated, and assessed using standard classification metrics (Accuracy, Precision, Recall, and F1-Score).

What Went Well:

- **Clean Model Integration:** Scikit-learn pipelines streamlined preprocessing (e.g., imputation, scaling), model training, and prediction, resulting in efficient and repeatable workflows.
- **Perfect Model Accuracy:** Both models achieved 100% performance on the test set and under 5-fold cross-validation, indicating a well-aligned dataset and clear separation between classes.
- **Interpretability with Logistic Regression:** Coefficients from Logistic Regression enabled direct insights into how features like retention rate (RET_FT4) and faculty salary (AVGFAC SAL) influenced risk.
- **Efficient Validation:** Cross-validation and confusion matrices helped confirm model stability and verify zero misclassification for both models.

What Could Be Improved:

- **Limited Feature Engineering:** Most features were used in raw form. Future iterations could include binning, polynomial interactions, or derived metrics (e.g., student support index).
- **No External Testing:** Model performance was only assessed on internal data. Future work should include external validation or temporal validation (e.g., different academic years).
- **Decision Tree Flexibility:** Although the Decision Tree performed perfectly, it used a shallow depth (max_depth=3). Exploring deeper trees or pruning strategies could better reveal its generalization capacity.
- **Other Classifiers Not Explored:** Models like Random Forest or Gradient Boosting were not tested due to scope limits but may offer improvements with more complex datasets.

Lessons Learned:

- **Model Transparency Matters:** Logistic Regression was favored due to its policy-aligned interpretability - ideal for communication with education stakeholders and administrators.
- **Modular Pipelines Are Critical:** Building models within reusable pipelines helped maintain consistency, simplify tuning, and streamline the comparison process.

- Default Models Can Perform Exceptionally: Even without hyperparameter tuning, default configurations in Scikit-learn produced accurate and stable results - reinforcing the importance of clean data preparation and proper variable selection.

Task 5.3 – Determining the Next Steps

Possible Actions

- Deploy the Logistic Regression model into a scoring environment to begin live risk predictions.
- Schedule periodic model monitoring to detect performance drift and recalibrate if accuracy declines.
- Store models output in a structured format for future auditability and future Phase 6.0 deployment tasks.
- Explore secondary models (e.g., Random Forest) in the future as a benchmark option if business needs evolve.

Final Decision

The final approved model for deployment is Logistic Regression, based on its stronger generalization performance, interpretability, and alignment with the original business goal which is to identify academically at-risk students early enough for intervention.

PHASE 6.0: Data Deployment

Task 6.1 Deployment Plan

Step	Task	Responsible	Start Date	End Date	Notes
1	Finalize logistic regression model code and export	Data Science Team	Dec 7, 2025	Dec 8, 2025	Ensure all preprocessing steps are integrated
2	Set up scoring environment in production (e.g., Flask API or cloud platform)	Data Engineer	Dec 9, 2025	Dec 11, 2025	Scoring endpoints will accept input features and return risk prediction
3	Deploy model in scoring environment	Data Engineer	Dec 12, 2025	Dec 12, 2025	Initial testing on test data
4	Conduct user acceptance testing (UAT)	QA / End users	Dec 13, 2025	Dec 15, 2025	Use a sample of known students
5	Go live with prediction dashboard/reporting	DevOps / Data Science	Dec 16, 2025	Dec 17, 2025	Stakeholders get access to weekly predictions
6	Store prediction logs for auditability	DevOps	Ongoing	Weekly	Log results in cloud-based database (e.g., AWS S3, Azure Blob)

To ensure the practical application of the predictive model, a comprehensive deployment plan was developed to outline how the final logistic regression model will be integrated into the institution's student support framework.

Task 6.2 Monitoring and Maintenance Plan

Monitoring & Maintenance Plan

Component	Metric	Frequency	Trigger for Action	Action to Take
Model accuracy	Accuracy on validation dataset	Monthly	Accuracy drops below 85%	Retrain model with recent data
Data drift	Feature distribution comparison (e.g., PSI or KS test)	Monthly	PSI > 0.2	Investigate data pipeline / update model
Prediction volume	# of predictions generated	Weekly	Sudden drop in volume	Check input pipeline or scoring environment

Component	Metric	Frequency	Trigger for Action	Action to Take
Error rate	% of missing or failed predictions	Daily automated alerts	>2% missing predictions	Dev team investigates scoring issue
Retraining schedule	Model retraining	Every 6 months	Scheduled	Retrain with latest academic year data

With monitoring intervals, maintenance responsibilities, and performance thresholds defined, the model is prepared for sustained operation. The following section presents the final results of the project and its impact on the business problem.

Task 6.3 – Reporting Final Results

Model Summary

The final model used for this project was a Logistic Regression classifier aimed at predicting students at risk of academic failure or dropout. The binary target variable was “At Risk” status, where 1 represented students flagged as high-risk and 0 indicated lower risk. Key features used in the model included GPA, enrollment intensity, Pell Grant recipient status, and tuition cost.

The model output a probability score between 0 and 1 for each student. A threshold of 0.50 was applied, classifying any student with a predicted probability greater than or equal to 0.50 as “at-risk.”

Key Findings

- Part-time enrollment and lower GPA were the strongest indicators of academic risk.
- Pell Grant recipients showed a higher likelihood of being flagged, suggesting financial support recipients may face more retention barriers.
- Tuition cost had a meaningful impact: students paying high tuition out-of-pocket were less likely to be flagged compared to those on financial aid, implying complex socioeconomic dynamics.

Model Performance

- The logistic regression model showed acceptable classification performance. Evaluation metrics such as accuracy, precision, recall, and F1-score indicated that the model was effective in identifying at-risk students.

- Confusion matrix analysis showed a balanced tradeoff between true positives (correctly identified at-risk students) and false positives, supporting its reliability for early intervention use.
- Additional modeling (e.g., decision trees) was considered, but logistic regression was selected for its interpretability and transparency, which is ideal for communicating results to academic advisors and administrators.

Business Impact

The model addresses a critical business need: improving student retention and reducing dropout rates. The impact is summarized as follows:

- Enables proactive intervention (tutoring, counseling, academic support) for high-risk students.
- Supports strategic resource allocation, such as targeting scholarships, mentorships, or emergency aid.
- Improves long-term outcomes by potentially increasing graduation rates and reducing tuition revenue loss.

Alignment with Project Goals

The project successfully met its original goal of developing a data-driven solution for early identification of at-risk students using institutional data. The model produced actionable insights that align with both academic and administrative objectives, demonstrating the value of predictive analytics in higher education

Task 6.4: Review Project

Team Experience Assessment

Although this CS703 project was structured to support team collaboration, I completed the entire lifecycle of the project independently. As such, my reflections focus on the experience of managing a full data science project solo, including planning, execution, evaluation, and reporting.

1. Collaboration Dynamics and Communication

Since there was no team assigned for this project, there were no inter-team communications or collaboration dynamics to reflect on. However, I did maintain regular contact with my professor to seek clarification and ensure alignment with CRISP-DM requirements. This guidance helped ensure that my independent work met the expectations of each deliverable.

2. Role Management and Task Distribution

In a typical team setting, roles such as data engineer, modeler, and business analyst might be divided among team members. In this project, I assumed all roles and was solely responsible for:

- Data acquisition and cleaning
- Exploratory analysis and feature engineering
- Model building and evaluation
- Documentation and final reporting

This required strong time management, organization, and adaptability to transition between tasks typically assigned to multiple roles.

3. Strengths and Challenges

Strengths:

- Completing the entire project independently strengthened my technical confidence, especially in data preprocessing, model implementation, and evaluation.
- I was able to maintain consistency in writing style, analysis depth, and overall presentation since every aspect was handled by me.
- The decision-making process was efficient, allowing for flexibility in selecting modeling techniques and project direction.

Challenges:

- The absence of team brainstorming meant all decisions were made independently, which could potentially limit the exploration of alternative strategies or model enhancements.
- Managing multiple deliverables while also balancing academic, professional, and personal commitments required discipline and careful planning.

4. Lessons Learned and Future Improvements

This solo project experience reinforced the importance of following a structured methodology like CRISP-DM. It also highlighted the critical role of documentation and version control in complex projects. In future team-based projects, I will apply the same level of structure and clarity, while also incorporating collaborative tools like Git for shared codebases and Trello or Asana for task tracking.

5. Overall Reflection

While the original intent of this deliverable was to evaluate team dynamics, my individual journey through this project offered a unique opportunity to simulate a full-scale data science workflow from end to end. It served as a capstone experience, consolidating the skills and knowledge acquired throughout the Master of Science in Data Science program.

Conclusion

This project successfully applied the CRISP-DM framework to develop and evaluate a classification model aimed at identifying colleges and universities with elevated academic risks.

Using an integrated dataset combining College Scorecard and IPEDS data, the analysis addressed clearly defined business and data mining goals. After a thorough comparison of multiple algorithms, Logistic Regression was selected for its strong performance and interpretability - both critical in academic settings where transparency and actionable insight are essential. Evaluation metrics and confusion matrix results confirmed the model's reliability, supporting its readiness for deployment. A detailed plan was also established to guide ongoing monitoring and future retraining.

Ultimately, this project demonstrated the practical value of predictive analytics in supporting student success and enabling institutions to take proactive steps toward academic improvement. All required deliverables were completed and documented, and the project was successfully completed.